Lagrangians

ex: $\mathcal{L} = \frac{1}{2} m (v_x^2 + v_y^2)$

switch to (r, θ) as coords

$x = r\cos\theta$ $\qquad$ $\dot{x} = \dot{r}\cos\theta - r\sin\theta\,\dot{\theta}$

$y = r\sin\theta$ $\qquad$ $\dot{y} = \dot{r}\sin\theta + r\cos\theta\,\dot{\theta}$

$\dot{x}^2 + \dot{y}^2 = \dot{r}^2 + r^2\dot{\theta}^2$ $\qquad$ $\mathcal{L} = \frac{1}{2} m (v_r^2 + r^2 v_\theta^2)$

EL

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial v_x} - \frac{\partial \mathcal{L}}{\partial x} = 0 \quad \rightarrow \quad \ddot{x} = 0$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial v_y} - \frac{\partial \mathcal{L}}{\partial y} = 0 \quad \rightarrow \quad \ddot{y} = 0$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial v_r}(r(t), \theta(t), \dot{r}(t), \dot{\theta}(t)) - \frac{\partial \mathcal{L}}{\partial r}(r(t), \theta(t), \dot{r}(t), \dot{\theta}(t)) \rightarrow$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial v_\theta}(r(t), \theta(t), \dot{r}(t), \dot{\theta}(t)) - \frac{\partial \mathcal{L}}{\partial \theta}(r(t), \theta(t), \dot{r}(t), \dot{\theta}(t)) \rightarrow$$

$$\frac{\partial \mathcal{L}}{\partial v_r} = m v_r \qquad \frac{\partial \mathcal{L}}{\partial v_\theta} = m r^2 v_\theta \qquad \frac{\partial \mathcal{L}}{\partial r} = m r v_\theta^2$$

$$\frac{d}{dt}(m\dot{r}) - m r \dot{\theta}^2 = 0 \quad \Big\| \quad \frac{d}{dt}(m r^2 \dot{\theta}) = 0$$

$\rightarrow$ conservation of angular momentum

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minimization problem

- in calculus if $f: U \subseteq \mathbb{R} \to \mathbb{R}$, the minima/maxima of f satisfy $f'(x) = 0$

generalize to functionals (functions of functions)

$L(\gamma)$ length of curve $\gamma: [0,1] \to \mathbb{R}^2$

in Euclidean space $\quad L(\gamma) = \int_0^1 \|\dot{\gamma}(t)\|\, dt$

Lagrangian

action

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solve using variations

back to calculus

a variation of the point $x \in \mathbb{R}$ is a function $\varepsilon \in (-1,1) \mapsto x_\varepsilon \in \mathbb{R}$

s.t. $x_0 = x$

$f'(x) = 0 \iff \frac{d}{d\varepsilon} f(x_\varepsilon)\Big|_{\varepsilon=0} = 0$ for all variations x_ε

$f'(x_0) \frac{dx_\varepsilon}{d\varepsilon}\Big|_{\varepsilon=0} = 0$ for all variations

$\iff f'(x_0) = 0$

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generalize if S is a functional on some space of functions $\{\gamma : [t_0, t_1] \to M\}$

variation of γ: $(\varepsilon, t) \in (-1,1) \times [t_0, t_1] \mapsto \gamma_\varepsilon(t) \in M$ (homotopy)

can also think of γ_ε as a curve for each ε

require $\gamma_0 = \gamma$.

γ is a critical point of S if $\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} S(\gamma_\varepsilon) = 0$ for all variations of γ

ex $\gamma : [0,1] \to \mathbb{R}$ $\quad S(\gamma) = \int_0^1 \gamma(t)\, dt$

γ_ε arbitrary variation $\quad \frac{d}{d\varepsilon} S(\gamma_\varepsilon)\Big|_{\varepsilon=0} = \int_0^1 \frac{d}{d\varepsilon}\gamma_\varepsilon(t)\Big|_{\varepsilon=0} dt$ not always 0

$\to$ arbitrary

no critical points

$S(\gamma) = \int_0^1 \gamma(t)^2\, dt$

$\frac{d}{d\varepsilon} S(\gamma_\varepsilon)\Big|_{\varepsilon=0} = \int_0^1 \frac{d}{d\varepsilon}\gamma_\varepsilon^2(t)\Big|_{\varepsilon=0} dt = \int_0^1 2\gamma(t) \frac{d}{d\varepsilon}\gamma_\varepsilon(t)\Big|_{\varepsilon=0} dt = 0$ for all variations

if $\gamma(t) = 0 \ \forall t$

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fix endpoints: $\gamma : [t_0, t_1] \to \mathbb{R}$ $\quad$ fix $\gamma(t_0) = a$ $\quad \gamma(t_1) = b$

variation with fixed endpoints: $\gamma_\varepsilon(t_0) = \gamma(t_0)$ $\quad \gamma_\varepsilon(t_1) = \gamma(t_1) \ \forall \varepsilon$

A special subclass of variational problems look like this:

$$S(\gamma) = \int_{t_0}^{t_1} \mathcal{L}(t, \gamma(t), \dot\gamma(t), \ddot\gamma(t), \dots)\, dt \qquad \gamma : [t_0, t_1] \to \mathbb{R}^n$$

$$\mathcal{L} : \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^n \times \dots \times \mathbb{R}^n \to \mathbb{R} \qquad \text{(Lagrangian)}$$

in physics generally we stop at 1 derivative

<u>Hamilton's principle:</u> the physical path is a critical point of S.

<u>Euler Lagrange eqs:</u> $\frac{d}{d\varepsilon} S(\gamma_\varepsilon)\Big|_{\varepsilon=0} = 0 \quad \forall$ variations

$$\mathcal{L} = \mathcal{L}(t, q, v) \qquad q \in \mathbb{R}^n$$

$$S(\gamma_\varepsilon) = \int_{t_0}^{t_1} \mathcal{L}(t, \gamma_\varepsilon(t), \dot\gamma_\varepsilon(t))\, dt$$

$$\frac{d}{d\varepsilon} S(\gamma_\varepsilon)\Big|_{\varepsilon=0} = \int_{t_0}^{t_1} \frac{d}{d\varepsilon} \mathcal{L}(t, \gamma_\varepsilon(t), \dot\gamma_\varepsilon(t))\Big|_{\varepsilon=0} dt$$

$$= \int_{t_0}^{t_1} \left[\frac{\partial \mathcal{L}}{\partial q^\mu}(t, \gamma(t), \dot\gamma(t)) \frac{d}{d\varepsilon} \gamma_\varepsilon^\mu(t)\Big|_{\varepsilon=0} + \frac{\partial \mathcal{L}}{\partial v^\mu}(t, \gamma(t), \dot\gamma(t)) \frac{d\, \dot\gamma_\varepsilon^\mu(t)}{d\varepsilon}\Big|_{\varepsilon=0} \right] dt$$

$$\frac{d\, \dot\gamma_\varepsilon^\mu(t)}{d\varepsilon}\Big|_{\varepsilon=0} = \frac{d}{dt}\left(\frac{d}{d\varepsilon} \gamma^\mu(t) \right)\Big|_{\varepsilon=0}$$

integration by parts

$$\frac{d}{d\varepsilon} \gamma_\varepsilon^\mu(t) = \delta q^\mu(t)$$

$$= \int_{t_0}^{t_1} \left[\frac{\partial \mathcal{L}}{\partial q^\mu}(t, \gamma(t), \dot\gamma(t))\, \delta q^\mu(t) - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial v^\mu}(t, \gamma(t), \dot\gamma(t)) \right) \delta q^\mu(t) \right] + \left[\frac{\partial \mathcal{L}}{\partial v^\mu}(t, \gamma(t), \dot\gamma(t))\, \delta q^\mu(t) \right]_{t_0}^{t_1}$$

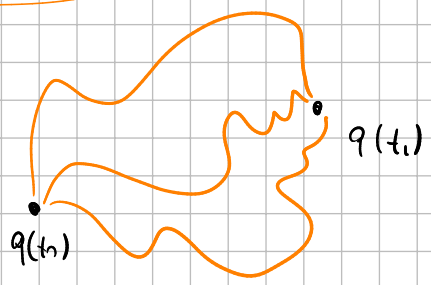
① γ is a critical point, then $\frac{d}{d\varepsilon} S(\gamma_\varepsilon)\Big|_{\varepsilon=0} = 0$ for all variations

$\Rightarrow$ so it's zero for all variations with fixed end points

so $\left[\frac{\partial \mathcal{L}}{\partial v^\mu}(t, \gamma(t), \dot\gamma(t))\, \delta q^\mu(t) \right]_{t_0}^{t_1} = 0$

so $\displaystyle\int_{t_0}^{t_1} \left[\frac{\partial \mathcal{L}}{\partial q^\mu}(t,\gamma(t),\dot\gamma(t))\, \delta q^\mu(t) - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial s^\mu}(t,\gamma(t),\dot\gamma(t)) \right) \delta q^\mu(t) \right] dt = 0 \qquad \forall\ \delta q^\mu$

$$\Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial q^\mu}(t,\gamma(t),\dot\gamma(t)) - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial s^\mu}(t,\gamma(t),\dot\gamma(t)) \right) = 0}$$



<u>general form</u>: $\mathcal{L} = \mathcal{L}(t, q_0, q_1, q_2, q_3, \dots, q_n)$

$$S = \int_{t_0}^{t_1} \mathcal{L}(t, \gamma(t), \dot\gamma(t), \ddot\gamma(t), \dots)\, dt$$

<u>EL</u>: $\boxed{\displaystyle\sum_{k=0}^{n} (-1)^k \frac{d^k}{dt^k} \frac{\partial \mathcal{L}}{\partial q_k^\mu}(t,\gamma(t),\dot\gamma(t),\dots) = 0}$

<u>$n=2$</u> $\displaystyle\frac{\partial \mathcal{L}}{\partial q^0} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial q^1} + \frac{d^2}{dt^2}\frac{\partial \mathcal{L}}{\partial q^2} = 0$

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adding a <u>total derivative</u> to the Lagrangian does not change the result

often physicists write $\mathcal{L} = \frac{1}{2} m \dot x^2 - \frac{1}{2} k x^2$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot x} - \frac{\partial \mathcal{L}}{\partial x} = 0$$

<u>proper way</u>: $\mathcal{L} = \frac{1}{2} m s^2 - \frac{1}{2} k x^2 \qquad \mathcal{L}: \mathbb{R}^2 \to \mathbb{R}$, $(x,s) \mapsto \mathcal{L}(x,s)$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial s}(x(t), \dot x(t)) \right) - \frac{\partial \mathcal{L}}{\partial x}(x(t), \dot x(t))$$

$$\frac{d}{dt}(m\dot x) + k x = 0 \quad \Rightarrow \quad m\ddot x = -k x$$

if $\mathcal{L} \to \mathcal{L} + \frac{d}{dt}(f(x,\dot{x}))$ then same equations (what physicist say)

what they mean: if you add $F(x,v,a)$ to $\mathcal{L}$

s.t. $F(x(t), \dot{x}(t), \ddot{x}) = \frac{d}{dt}\left(f(x(t), \dot{x}(t))\right)$ then EL eq. are the same

$$\tilde{\mathcal{L}}(t,q,v,a) = \mathcal{L}(t,q,v) + \frac{\partial f}{\partial q} v + \frac{\partial f}{\partial v} a \qquad \text{where } f = f(q,v)$$

<u>EL:</u>
$$\frac{\partial \tilde{\mathcal{L}}}{\partial q}(t,x,\dot{x},\ddot{x}) - \frac{d}{dt}\left(\frac{\partial \tilde{\mathcal{L}}}{\partial v}(t,x,\dot{x},\ddot{x})\right) + \frac{d^2}{dt^2}\left(\frac{\partial \tilde{\mathcal{L}}}{\partial a}(t,x,\dot{x},\ddot{x})\right) = 0$$

$$\frac{\partial \mathcal{L}}{\partial q} + \cancel{\frac{\partial^2 f}{\partial q^2}\dot{x}} + \cancel{\frac{\partial^2 f}{\partial q \partial v}\ddot{x}} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial v} + \cancel{\frac{\partial^2 f}{\partial q \partial v} v} + \cancel{\frac{\partial f}{\partial q}} + \cancel{\frac{\partial^2 f}{\partial v^2} a}\right)$$
$$+ \frac{d^2}{dt^2}\left(\cancel{\frac{\partial f}{\partial v}}\right)$$

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geometrically $\mathcal{L} = \mathcal{L}(q,v)$ $\quad q \in M$ manifold

$v \in T_q M$

$\mathcal{L} : TM \to \mathbb{R}$

$\ddot{x} = 0 \iff \begin{array}{l} \dot{v} = 0 \\ \dot{x} = v \end{array}$

$M = \mathbb{R}^2$ $\qquad \mathcal{L}(q,v) = \frac{1}{2} m \|v\|^2$

$v = v_x \frac{\partial}{\partial x} + v_y \frac{\partial}{\partial y}$ $\qquad \|v\|^2 = v_x^2 + v_y^2$ $\qquad \left\langle \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right\rangle = \delta_{ij}$

<u>change coordinates:</u> r, θ $\qquad x = r\cos\theta$, $y = r\sin\theta$

$dx = dr\cos\theta - r\sin\theta\, d\theta$

$dy = dr\sin\theta + r\cos\theta\, d\theta$

$$\frac{\partial}{\partial r} = \frac{\partial x}{\partial r}\frac{\partial}{\partial x} + \frac{\partial y}{\partial r}\frac{\partial}{\partial y}$$

$$\frac{\partial}{\partial \theta} = \frac{\partial x}{\partial \theta}\frac{\partial}{\partial x} + \frac{\partial y}{\partial \theta}\frac{\partial}{\partial y}$$

$$\frac{\partial}{\partial r} = \cos\theta \frac{\partial}{\partial x} + \sin\theta \frac{\partial}{\partial y} \qquad v = v_r \frac{\partial}{\partial r} + v_\theta \frac{\partial}{\partial \theta}$$

$$\frac{\partial}{\partial \theta} = -r\sin\theta \frac{\partial}{\partial x} + r\cos\theta \frac{\partial}{\partial y} \qquad \|v\|^2 = ?$$

$$\left\langle \frac{\partial}{\partial r}, \frac{\partial}{\partial r} \right\rangle = \left\langle \cos\theta \frac{\partial}{\partial x} + \sin\theta \frac{\partial}{\partial y}, \cos\theta \frac{\partial}{\partial x} + \sin\theta \frac{\partial}{\partial y} \right\rangle$$

$$= (\cos\theta)^2 + (\sin\theta)^2 = 1$$

$$\left\langle \frac{\partial}{\partial r}, \frac{\partial}{\partial \theta} \right\rangle = \left\langle \cos\theta \frac{\partial}{\partial x} + \sin\theta \frac{\partial}{\partial y}, -r\sin\theta \frac{\partial}{\partial x} + r\cos\theta \frac{\partial}{\partial y} \right\rangle$$

$$= -r\cos\theta \sin\theta + r\cos\theta \sin\theta = 0$$

$$\left\langle \frac{\partial}{\partial \theta}, \frac{\partial}{\partial \theta} \right\rangle = \left\langle -r\sin\theta \frac{\partial}{\partial x} + r\cos\theta \frac{\partial}{\partial y}, -r\sin\theta \frac{\partial}{\partial x} + r\cos\theta \frac{\partial}{\partial y} \right\rangle$$

$$= r^2$$

$$v = v_r \frac{\partial}{\partial r} + v_\theta \frac{\partial}{\partial \theta}$$

$$\|v\|^2 = v_r^2 + r^2 v_\theta^2$$

<u>physicists do this</u>

$$\mathcal{L} = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) \qquad x = r\cos\theta, \quad y = r\sin\theta$$

$$\dot{x} = \dot{r}\cos\theta - r\sin\theta\,\dot{\theta}$$

$$\text{- - - -} \Rightarrow \mathcal{L} = \frac{1}{2} m (\dot{r}^2 + r^2\dot{\theta}^2)$$

<u>Mechanics</u> $\quad q$ position of a particle $\quad V(q)$ potential energy

v velocity of particle

$$\mathcal{L} = \frac{1}{2} m v^2 - V(q) \rightarrow \text{EL: } m\dot{v} = -\nabla V(q) = \text{force}$$

<u>for fields</u>

$$\mathcal{L} = \frac{1}{2} m \left(\frac{d}{dt} x\right)^2 - V(x) \qquad \| \quad x = x(t)$$

$$\downarrow \text{field}$$

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \, \partial^\mu \phi - V(\phi) \qquad \| \quad \phi = \phi(t, x)$$

$$\boxed{\frac{\partial}{\partial x^\mu}\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} - \frac{\partial \mathcal{L}}{\partial \phi} = 0}$$

$$\mathcal{L} = \frac{1}{2}\partial_\mu \phi \partial^\mu - V(\phi)$$

$$\frac{\partial}{\partial x^\mu}\frac{\partial \mathcal{L}}{\partial \partial_\mu}(\phi, \partial\phi) - \frac{\partial \mathcal{L}}{\partial \phi}(\phi, \partial\phi)$$

in Minkowski: $\partial_t^2 - \nabla^2$

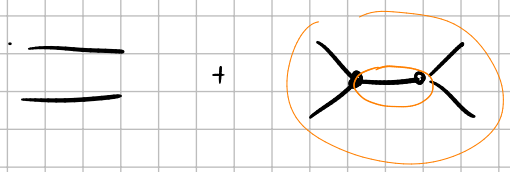
$$\rightarrow \quad \partial_\mu \partial^\mu \phi = -V'(\phi)$$

<u>in QFT</u> start with free Lagrangian for your kind of field

$$\mathcal{L}_F = \frac{1}{2}\left(\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2\right)$$

$$\mathcal{L}_F = -\frac{1}{4}\mathrm{tr}\left(F_{\mu\nu}F^{\mu\nu}\right)$$

<u>ex:</u> add ϕ^3 interaction: $\mathcal{L} = \frac{1}{2}\left(\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2\right) + g\phi^3$

canonical quantization $\phi(p) \rightarrow \hat{\phi}(p)$ (operator)

$$\langle 0|\phi(q)\,\phi(p)|0\rangle = \delta(p-q)\,\frac{1}{p^2} \quad \text{(free Lagrangian)}$$